

Multidetector Row Computed Tomography in the Management of Penetrating Neck Injuries

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Penetrating neck trauma may occur from gun shots, stabbings, and accidental injury. Approximately 50% of gunshot and 10%-20% of stab-wound patients are reported as having severe injuries and the mortality from severe vascular injuries is reported as high as 50%. Penetrating traumatic neck injury is no longer best managed with exploratory surgery or conventional angiography in the stable patient. Computed tomography angiography has proven to be a useful, safe, and reliable means of diagnosis. Experience with interpreting and reconstructing computed tomography images, understanding the clinically important findings, and avoiding pitfalls is critical for successful patient care. Therefore, radiologists and clinicians who treat trauma patients need to familiarize themselves with the computed tomography angiography technique and appearances of vascular injury and know when to recommend further evaluation, such as interventional angiography or open surgery. Semin Ultrasound CT MRI 30:195-204 © 2009 Elsevier Inc. All rights reserved.

Penetrating neck trauma may occur from gun shots, stabbings, and accidental injury. Approximately 50% of gunshot and 10%-20% of stab-wound patients are reported as having severe injuries¹⁻³ and the mortality from severe vascular injuries is reported as high as 50%.⁴ In our experience, however, the incidence of severe injury in the neck following penetrating trauma is much lower.

Managing penetrating neck trauma in the acute setting has been fraught with controversy. Between World War I and World War II, the preferred treatment changed from conservative observation to surgical exploration in an effort to decrease mortality rates. This held through the Vietnam era⁵ but as imaging technologies permitted less invasive evaluations the standard of care has slowly been changing to favor diagnostic imaging in hemodynamically stable patients.¹ While ultrasound and magnetic resonance angiography (MRA) have been studied in this setting, computed tomography angiography (CTA) is currently the most widely used and accepted technique. At most institutions CTA has replaced digital subtraction angiography as the initial study of choice for the evaluation of the neck vessels. Experience with interpreting and reconstructing computed tomographic (CT) images, understanding the clinically important findings, and avoiding pitfalls is critical for successful patient care.

Anatomical Considerations

The neck is subdivided into 3 anatomic zones based on access to underlying anatomy. Zone 1 encompasses the region from the clavicles and sternal notch to the cricothyroid membrane and is treated similarly to a thoracic injury. Zone 2 includes the area between the cricothyroid membrane and the angle of the mandible. Zone 3 includes the area from the angle of the mandible to the skull base (Fig. 1).⁶ Making clinical decisions based on these anatomic zones is problematic because an external wound to a particular zone may not reflect the trajectory of the knife or bullet. The zonal schema, however, is still referenced in clinical discussions.⁷

The traditional spaces of the neck used in locating masses have not found widespread utility in the trauma setting. Understanding the fascial planes may prove useful in locating hematomas and metallic fragments that suggest injury to a particular structure. The carotid space contains the carotid artery, internal jugular vein, and nerves. Evidence of hemorrhage clearly within the carotid space warrants close clinical follow-up and should raise the level of suspicion for the presence of a vascular injury. The retropharyngeal space is a potential space posterior to the pharynx, anterior to the prevertebral musculature, and is most important in the trauma setting because a retropharyngeal carotid artery may be identified. This variant anatomy is important to recognize for presurgical planning and as a warning to an aggressive esophagoscopist looking for esophageal injury. The perivertebral space includes the muscles and soft tissues around the

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